[65] A. R. Padhani, K. A. Lecouvet, and N. Tunariu, "MRI in prostate cancer: Functional imaging and future directions," Eur. Radiol., vol. 19, no. 4, pp. 1157–1168, Apr. 2009.

[66] D. Ravi et al., "Deep learning for health informatics," IEEE J. Biomed. Health Inform., vol. 21, no. 1, pp. 4–27, Jan. 2019.

[67] L. Zhou, Y. Wang, and L. Wang, "Alzheimer's disease detection through multimodal neuroimaging with CNNs," Neurocomputing, vol. 402, pp. 121–128, Apr. 2020.

[68] L. Perez and J. Wang, "The effectiveness of data augmentation in image classification using deep learning," arXiv preprint arXiv:1712.04621, Dec. 2017.

[69] A. Esteva et al., "Dermatologist-level classification of skin cancer with deep neural networks," Nature, vol. 542, no. 7639, pp. 115–118, Feb. 2017.

[70] [A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in Advances in Neural Information Processing Systems, 2012, pp. 1097–1105.

[71] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248–255.

[72] P. Rajpurkar et al., "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," arXiv preprint arXiv:1711.05225, 2017.

[73] M. A. Brahim et al., "Transfer learning for MRI brain tumor classification: A comparative study," Biocybernetics and Biomedical Engineering, vol. 41, no. 3, pp. 1024–1041, 2021.

[74] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," arXiv preprint arXiv:1409.1556, 2014.

[75] J. Irvin et al., "CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, no. 01, pp. 590–597, 2019.

[76] K. He et al., "Deep residual learning for image recognition," in Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 770–778.

AI Powered Snake Bite Classification System Using MobileNetV2 for Venomous Snake Identification

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***Abstract*— Snake bite incidences in Sri Lanka, especially in rural agricultural areas are, one of the critical public health issues with more than 80,000 bite incidences every year of which at least 400-600 are fatal. The identification of snake species needs to be done quickly so that the proper antivenom can be administered to the patient to save the life. The study is to develop SerpoDetect, a mobile app to identify snakes using the bite marks and symptoms using AI and deep learning. The tool aims to be lightweight without sacrificing accuracy and is meant for all people with smart devices to access. The convolutional neural network MobileNetV2 was selected for efficiency and scalability. To get significant amounts of training data, a dataset of 125 annotated snakebite images created via preprocessing techniques like data augmentation was compiled. The model performed training and validation, obtaining an impressive accuracy of 97.7% and an F1 score of 0.98 and 0.97 with regard to venomous and non-venomous snake classifications, respectively. The dual-input SerpoDetect enables you to upload a photo of your snakebite along with symptoms that make its diagnosis more productive by comparing visuals with symptomatic input. It includes such minor elements such as first aid instructions and emergency contact numbers in order to improve both standards of medical personnel and layman patients in remote areas. It shows that these AI tools have the potential to speed up snakebite identification and enhance accuracy in diagnosis and treatment, thereby reducing mortality. Future work will put efforts towards expanding the dataset, improving real-time image analyses, and to facilitate accessibility by more users. It transforms the nature of the healthcare problems by machine-learning solutions that are now in serious debate.**

Keywords—Snakebite, machine learning, deep learning, MobileNetV2, AI-driven application, symptom analysis, snake identification.

I. INTRODUCTION

A. *Background*

Snakebite has become a concern viewed from the public health lens of Sri Lanka, especially in rural agricultural areas because of the common encounters with such venomous snakes. Annually, about 80,000 snakebite cases are reported leading to over 30,000 hospital admissions and about an estimated 400-600 deaths [10, 6]. However, these totals are not accurate absolute figures, as there is a significant unreported figure due to reliance on traditional medicine and remoteness that limits access to healthcare facilities. To select the appropriate antivenom administered, the species of snake causing the bite must usually be accurately identified. Conventional identification methods unfortunately often lack dependability and consequently result in delays and inaccuracies in treatment. Progress in machine learning as well as image processing indicates considerable promise in automating snakebite mark analysis for faster and more accurate diagnosis, thereby potentially reducing snakebite mortality [2].

B. *Problem Statement*

The problem with snakebites is identifying if the snake is venomous or non-venomous. If survivors know the family of the snakes, then easily doctors will take perfect anti-venom and clinical procedures to cure the patient. But if the snake's family is unknown, it becomes harder for doctors to treat the patient. Based on this, ongoing research work is being conducted in this area to improve identification of snakes which bite the patient to help doctors select an appropriate anti-venom.

C. *Justification*

Sri Lanka's countryside agricultural areas face severe dangers of snakebites as one of the major public health issues. The identification of snake species is essential for providing the correct treatment, traditional identification methods are often unreliable and can make late treatment. By implementing machine learning techniques for the analysis of snake bite marks and symptoms, this research focuses on providing a faster and more reliable alternative. The development of a machine learning-based identification system generally has an effect on increasing the diagnostic accuracy of medical practices, saving lives, and reducing healthcare demands, especially within areas that lack the necessary medical resources [10, 6].

D. *Scope and Limitations*

The present work encompasses a study to develop a model using machine learning effectively to identify snake species accurately-from their bite marks and symptoms. It collects and analyzes an array of bite images, trains the models, and creates a user interface for health professionals. However, there were some limitations, such as image quality, a smaller dataset, and some regions with different snake species, which could cause the model to be inaccurate. Nevertheless, this research has the potential to yield significant improvements in snakebite management, especially in resource-poor parts of the world.

II. LITERATURE REVIEW

A. *Research Theme*

Snake bites pose serious medical hazards, especially in areas where venomous snake species are more abundant. Proper identification of the snake species is the first step in giving the right type of antivenom. Traditionally, the snake is identified by doctors according to the symptoms. Machine learning and image processing have worked out better ways of identifying snakes better beyond using symptoms. This review examines the contemporary studies of the bite mark

images and symptoms used in differentiating venomous and non-venomous snakes.

B. Theoritical Explanation about the Key Concepts

- Image Processing: The process of image processing involves applying algorithms to improve the quality of images or extracting relevant features from them. Typical methods include edge detection, contour detection, and feature extraction.
- Machine Learning: It is the training of algorithms to forecast results based on available data. In the context of snake bite identification, it makes use of the image features to classify the bite marks as venomous or nonvenomous in effective manner.
- Deep Learning: A type of machine learning that uses several different types of neural networks for identifying more complex patterns. CNNs have long been and are still being used to analyze pictures., including snake bite marks.
- Venomous vs Non-venomous Snakes: Venomous snakes will normally inject toxins which will give symptoms like pain and swelling while non-venomous snakes will comparatively have milder effects.
- Snakes Bite Marks: Snakes can bite humans or animals. Venomous snakes bite their prey and inject venom from their glands. The venom can poison the prey, causing paralysis or death quickly. When a snake bites, it's important to check the bite marks. The bite marks can help identify if the snake is venomous or non-venomous.

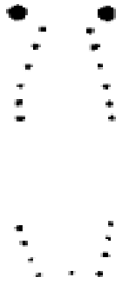


Fig. 1. Venomous Snake Bite Marks

As shown in **Fig 1**, the upper two dark black dots in the illustration, Figure 1, represent the marks made by the fang of venomous snakes. The smaller dots around them represents other marks made by the snake's teeth.



Fig. 2. Non-Venomous Snake Bite Marks

The bite marks shown above are common for non-venomous snakes. However, some non-venomous snakes can also keep fang marks. Their bite marks usually look like an 'arc' with some of closely spaced dots. These marks are easy for survivors to see and help them identify if the snake was venomous or non-venomous.

C. Findings

Numerous researchers have delved into the machine learning techniques that are employed for image classification with the distinct interest on some methods like K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Convolutional Neural Networks (CNN) in making the right predictions.

- K-Nearest Neighbors (KNN): KNN (K-nearest neighbor) is a model that classifies data based on the distance from labelled neighbors. KNN is a good model when it comes to simple classifications, but for high dimensional datasets such as snake bite images, it often does not reproduce the cases well, resulting in an over fit in most complicated situations [17].
- Support Vector Machine (SVM): SVM algorithms make use of hyperplanes to separate datasets into different classes. The performance is relatively robust, especially in tasks, such as distinguishing venomous bites from non-venomous ones. Nevertheless, it requires careful parameter tuning, otherwise, problems arise as the size of the dataset increases or becomes noisy [7].
- Convolutional Neural Networks (CNNs): CNNs basically function in a hierarchic way to process the photos, extracting the features from low level to high level. According to studies [7, 12], CNNs achieve better results than any other model in the classification of snake bites. However, to train a CNN, a huge amount of data must be fed in the form of labelled images, and usually, such data is not available. Transfer learning using pretrained models like MobileNetV2 tackles this problem and increases the accuracy while reducing the requirement of computation resources.

Instead of these advancements, there are several challenges,

- General limitations of dataset: Most studies are done under poorly labelled conditions or lack appropriately sized dataset [16].
- Species Misclassification: Most systems classify bitten victims as either venomous or non-venomous without giving the snake species [4].
- Irregularities in Skin Texture: Techniques like Local Binary Patterns in combination with SVM suffer in differentiating bite marks from skin irregularities [8].
- Integration Mismatches: Most systems do not use analysis symptoms connected to image classification, which reduces diagnostic accuracy [7].

D. *Gaps and Proposed Contributions*

The gaps that identified in existing research involves:

1) Limitation of dataset size and difference in labels.

2) Difficulty in differntiating actual bite marks from non-bite regions.

3) No integrated analysis of bite marks and symptoms.

The identified gaps target the proposed system as follows:

- Symptom analysis with image processing integration to increase the classification accuracy.
- Transfer learning utilization using MobileNetV2 to augment the available dataset and improve performance.
- A comprehensive approach of bite classification not limited to venomous and non-venomous snakes, but also classifying them into families.

III. METHODOLOGY

This research delves into image processing, machine learning, and deep learning to classify venomous and non-venomous snakes according to their bite marks and symptoms. Data collection, preprocessing, and analytical techniques comprise the methodology. The following stages lead to getting precise categorizations and support in timely administering the right anti-venom.

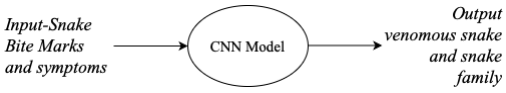
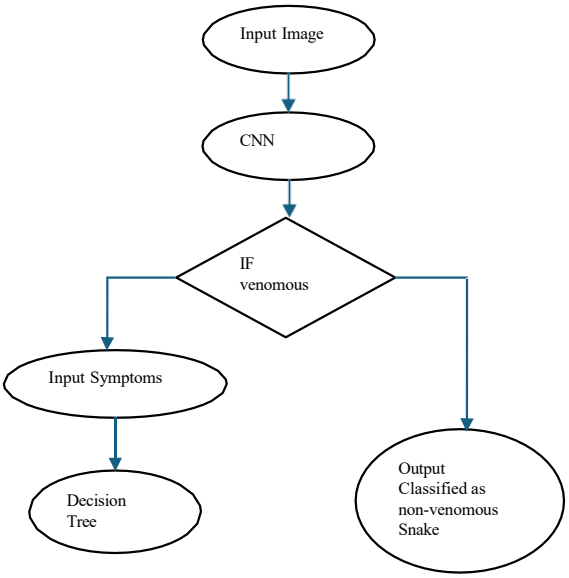


Fig. 3. Context Flow Model of Snake Identification

Below are the procedures of the identified CNN model for the identification of snakes.



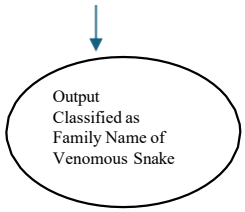


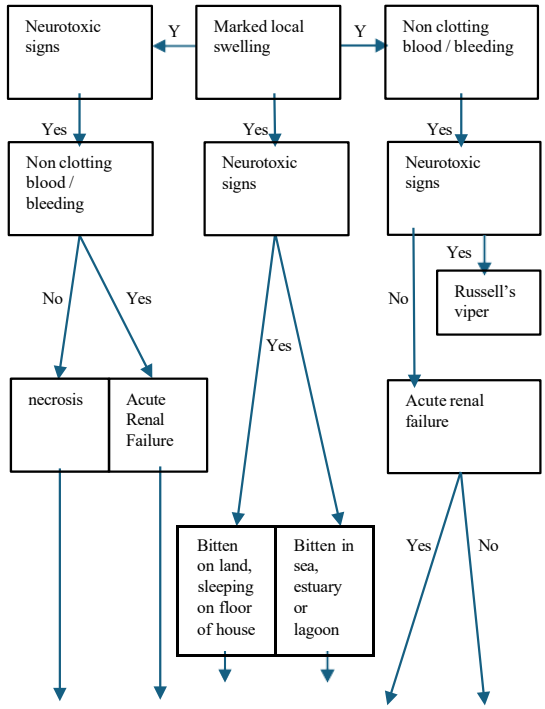
Fig. 4. Flow chart of the proposed system

1) *CNN*

The input images are taken to predict whether the snake bite is from a venomous or a non-venomous snake bites from a non-venomous snake are usually easy to read even with a little experience of such bite marks. However, even such bites should be captured by the way the system architectures the CNN itself because it then prevents the system from taking further unnecessary steps, thus saving energy and resources. If the bite is recognized as venomous snake, the result from CNN will be sent to the next step for further processing.

2) *Decision Tree*

The input is acquired from the CNN, which can identify the bitten marks of venomous snakes. And then, it works to determine the family of the snake based on the symptoms analysis using Decision tree algorithm. For this purpose, it compares the symptoms of the patient with those of different snake families. Once the matching has been done, the system gives an output of the result, including the family name of the snake. That helps decide future actions regarding patient treatment [18].



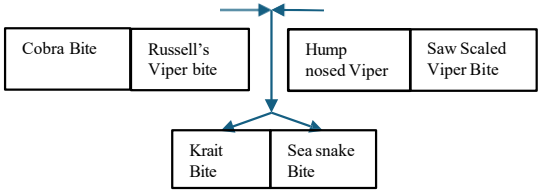


Fig. 5. Diagnosis of snakebite cases based on clinical data

The decision tree algorithm is helpful in analyzing the symptoms of a patient for bites and helps to find out which family of snakes it belongs to. This is designed to function like a flowchart and does symptoms structure into a hierarchy. The inputs are neurotoxic symptoms, local swelling, problems in clotting and renal failure. Based on the symptoms of the patient, the decision tree traverses different branches and hones down possible snake families. As an example, certain symptoms such as local swelling and neurotoxicity could be indicative of a cobra while, for example, spontaneous bleeding and acute renal failure might denote a viper. This structured approach would help in a faster, accurate decision-making process in patient care.

B. Population, Sample and Sampling Techniques

The study targets on the population of Northcentral Province in Sri Lanka, which records the highest snakebite incidence rate of 623 per 100,000 annually [6]. A secondary dataset of 125 snake bite images was collected. It included an equal representation of venomous and non-venomous cases. The data was gathered from medical universities and public health databases accordingly. Organized random sampling was used to make sure that venomous and non-venomous bites were illustrated on the right ratio. Secondary data became the primary concern due to the challenges associated with direct collection of qualitative images. Data sites were included as such health institutions and online database repositories.

TABLE IX. INCIDENCE OF SNAKEBITES IN SRILANKA

Province	Bites		
	Reported (sample)	*Estimated number*	*Risk per 100000 (95% CI)*
Western	61	18910	325 (217-432)
Central	45	7065	277(182-371)
Sabaragamuwa	102	10506	548(438-658)
Northwestern	92	11776	499(392-605)
Southern	87	11310	461(338-584)
Uva	63	4095	328 (242-414)
Nothern	59	3422	324 (219-428)
Eastern	67	5695	368 (227-509)
NorthCentral	119	7735	623(487-760)

[a.] Data involves estimated and reported snakebites cases, with risk per 100, 000 units and the corresponding 95% confidence interval.

[b.]

C. Data Collection and Sources

Information on snakebite photographs, symptoms, and treatment records like clinical procedures were collected from public health databases, medical records in hospitals and clinics, and recorded bite marks and patient outcomes. This would provide a well-rounded view of the snakebite incidence. This, however, has also confirmed the enriched and very illustrative dataset for analysis using multiple approaches.

D. Data Pre-Processing and Analysis

The data pre-processing and analysis techniques to enhance model performance through the delivery of accuracy outcomes would include data augmentation, normalization and good dataset.

Responding to the restriction of the available dataset size, data augmentation techniques were implemented, for flipping, rotation, cropping, and brightness adjustment. The images collected were then standardized to a size of 224 x 224 pixels in .jpeg format supported by the MobileNetV2 model. The dataset was divided by 70% for training and 30% for validation and then further classified into venomous and non-venomous subfolder which contains different snake bites. The entire structure made very easy processing and efficient model training. In addition to image classification, evaluation of symptoms was included to improve diagnostic accuracy reliably.

Critten, according to the established guidelines [13], compares clinical signs like neurotoxicity, local swelling, and blood clotting aspects. For instance, neurotoxic symptoms usually denote cobra or krait bites, whereas systemic bleeding indicates viper bites.

E. System Development

The development of the system was carried out using Python and deep learning frameworks such as TensorFlow and Keras on Google Colab making use of the GPU resources. Hardware requirements involved the use of 64-bit processor with 16GB RAM. The approach used in the development process was agile by utilizing an iterative and incremental way. This method allowed ongoing refinement of the snakebite classification model and refinement the mobile application interface. The model employed MobileNetV2 for snakebite image classification. Users upload images through the mobile application. The app processes the bite images with the trained model to provide detected snake type. Furthermore, Symptoms are analyzed to provide more diagnostic insights.

F. User Interface Design

As shown in **Fig. 10**, the user interface (UI) of the mobile application is designed to ensure ease of use and efficiency in identifying snake bites. The application consists of multiple screens, each serving a specific purpose to facilitate the user experience. The primary UI components include:

- **Login/Signup Screen**: Allows users to create an account or log in to access personalized features.
- **Home/Dashboard Screen**: Provides an overview of the application's functionalities, including bite image upload and symptom input.

- **Bite Image Upload Screen**: Enables users to upload an image of the snake bite, which will be analyzed by the machine learning model.
- **Symptom Input screen**: Allows users to enter symptoms they are experiencing, contributing to a more accurate identification of the snake type.
- **Analysis and Result Screen**: Displays the outcome of the system's analysis, indicating whether the snake is venomous and non-venomous.
- **Case History/Records Screen**: Maintains a history of previous diagnoses for future reference.
- **Emergency Contact Screen**: Provides quick access to emergency services and medical professionals.

The UI is designed to be intuitive, ensuring that users can easily navigate through the application without prior technical knowledge. The color scheme and layout have been chosen to enhance readability and accessibility.

G. *Sequence Diagram*

As shown in **Fig 11**, to illustrate the workflow of the mobile application, a sequence diagram is used to represent the interaction between users and the system. The diagram outlines the key processes involved in snake bite identification, showing the sequence of messages exchanged between different components.

- **User Logs In**: The user enters their credentials, and the system verifies authentication.
- **Image Upload and Symptom Entry**: The user uploads an image of the bite and enters symptoms.
- **Data processing**: The backend system processes the input using machine learning models.
- **Snake Identification**: The system predicts the type of snake based on the analysis.
- **Result Display**: The identified snake type and recommended actions are presented to the user.
- **Emergency Assistance**: The user can contact medical professionals through the emergency screen.

IV. DATA ANALYSIS

This work involved online secondary data. Specifically, snakebite images were sourced from the Internet showing distinguishable bite mark pattern on the individual bitten. MobileNetV2 is efficient lightweight model; it does best for application in case there is constraint in computational powers. It was used to process the images. The images were available in .jpeg and .arw formats and the images were converted to .jpg format for standardization. Low quality images such as blurred, very small ones were removed, and the existing images were manually cropped to emphasize bite marks clearly. This method decreased interruption from the background and enhanced the sharpness of the focused region. The images were consequently classified as poisonous and non-poisonous and resized to dimensions of 224x224 pixels to match the inputs needed for the model.

A. *Data Collection and Preparation*

In total, a collection of 125 photographs has been gathered from various sources, including medical records and online databases. After refined filtering, 87 pictures remained to qualify as high quality: 58 represent the number of venomous snake bites, while 29 are for bites from non-venomous snakes. Such a dataset confirmed balanced representation in both categories and opened avenues to good qualitative analysis. MobileNet was, thus, used for classifying the images by analyzing the unique visual features of bite patterns. Standard performance metrics such as accuracy, precision, recall, and F1 score were computed to judge the reliability of the model. The model achieved a fine accuracy of 97.7%. The F1 scores were 0.98 and 0.97 for venomous and non-venomous classification, respectively, with a recall of 1.00 for venomous cases perfectly. There was a clear prominence in the confusion matrix in limited counts showing misclassifications that resulted in only two venomous ones being classified as nonpoisonous.

B. *Finding and Interpretation*

Bite mark patterns can determine whether a snake bite is venomous or not. This report describes a deep learning model that could be harnessed in doing so. The system was capable of making accurate and robust generalization up to the level of consistent accuracy both for training and testing datasets. When comparing with previous works, for example, "Deep Learning Model for Identifying Snakes by using Snakes Bite Marks" [11], which gave an accuracy of 92%, and "A Deep Learning Approach for snake Bite Mark Identification and Classification" [9], which attained 85%, the proposed model got a remarkable improvement. This improvement was attributed to improved training procedures and modernized feature extraction methods. Such good performance results showed the capability of this method in practical scenarios in the clinical procedure of medical diagnosis snake bites.

V. DISCUSSION

This chapter shows the findings of the study and measures whether the research missions have been fulfil. The research concentrated on the challenges in developing a snake identification system and investigated machine learning techniques for image classification. Specially, the study searched how these methods can support in snake identification. After appraising several machine learning techniques, the research switch to transfer learning, with MobileNetV2, being selected due to its rightness for image processing tasks. As a result, mobile application integrated this deep convolutional neural network (CNN), designed to function efficiently on devices with small-scaled computational power.

A. *Validation of the Model*

In this research, transfer learning applies to a pretrained convolutional neural network (CNN) that is fine-tuned on snake bite classification tasks. The MobileNetV2 model has been chosen and modified to classify snake bites as venomous and non-venomous. The dataset of this research consisted of a total of 125 images collected from various online sites and hospital records according to the way that has been detailed in the methodology section. Seventy percent (70%) of these images were used for training, while thirty percent (30%) were assigned to validation and testing. The model's true performance evaluated using the test set,

which had not been detected by the model during training. The metrics like accuracy, precision, recall and support were utilized to evaluate the model clearly. The final accuracy was 97.7%, marking that the model had effectively absorbed from the training data.

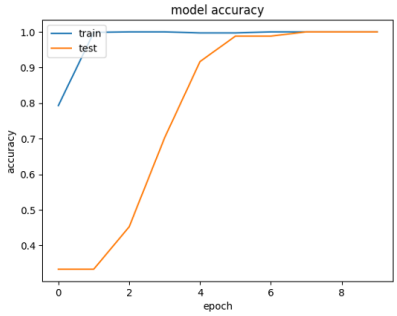


Fig. 6. Model Accuracy

As shown in **Fig 6**, shows the accuracy of the model during training and testing. The training accuracy started at a high level and quickly stabilized near 100%, while the testing accuracy initially increased gradually and reached approximately 100% after several epochs. This trend indicates that the model successfully learned patterns from the training data and generalized well to the test data.

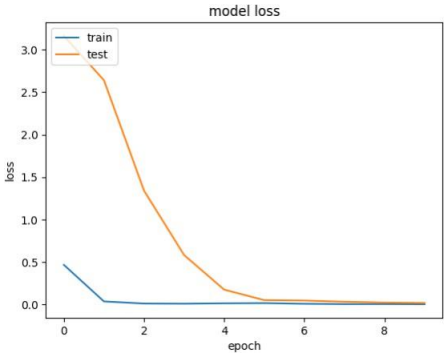


Fig. 7. Model Loss

As shown in **Fig 7,** presents the loss values during training and testing. The training loss decreased rapidly in the early epochs and remained close to zero, while the test loss initially had a high value but gradually reduced as the training progressed. The reduction in loss signifies that the model effectively minimized errors over time, leading to high performance.

As shown in **Fig 8,** confusion matrix displays how well the model performs regarding the classification of snakebites. It indicates that 56 venomous snakebite cases were identified correctly while 2 were misidentified. In addition, the 29 nonvenomous cases were also entirely identified without false negatives. This suggests very high accuracy for both venomous and nonvenomous classifications.

	precision	recall	f1-score	support
Venomous	1.00	0.97	0.98	58
Non_Venomous	0.94	1.00	0.97	29
accuracy			0.98	87
macro avg	0.97	0.98	0.97	87
weighted avg	0.98	0.98	0.98	87

Fig. 8. Classification Report

As shown in **Fig 9,** the high accuracy of 98% is due to several factors. The well-balanced dataset, with 58 venomous and 29 non-venomous samples, ensured the model learned both classes effectively. The deep learning design enabled precise feature extraction. This improved performance, especially for the venomous class, which reached perfect precision (1.00). Additionally, optimizing hyperparameters during training may have improved generalization. The low misclassification rate, shown by high recall (1.00) for non-venomous bites, shows effective handling of difficult cases.

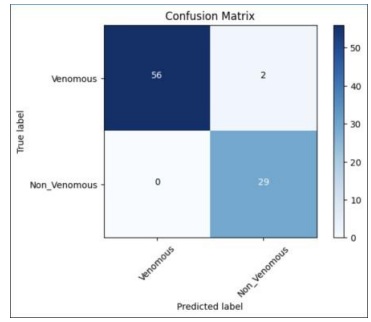


Fig. 9. Confusion Matrix

B. Achievements of the Objectives

The goal of this study was investigations into machine learning algorithms for image classification. Some of these algorithms include convolutional neural networks (CNNs) and transfer learning techniques. Models were trained in a snake bite image dataset with an emphasis on separate species classification. The metrics of evaluation include accuracy, precision, recall, and all such others which aided in the determination of the best models for the task concerned.

Secondly, it talked about the designing of a machine learning model which will be able to identify snake species automatically through their bite marks. As mentioned in the methodology chapter, the system was created to function without autonomously. The model was able to categorize snake bites as either venomous or non-venomous, an essential task given the direct link between the snake type and capable damage. Wrong identification of snake species can lead to significant medical and health implications so precision in this area becomes critical. The model given here has much improved over previous snake image classification models in accuracy. It was tested that MobileNetV2 could realize a better classification potential reaching the final testing accuracy of 96%, which was approved freely using

the data from the test set of unseen data. Evaluation measurement tools such as precision, recall, and F1 score were applied to test the validity of the model.

C. *Recommendatios for Future Work*

1) Improve the dataset to generalize better the model.

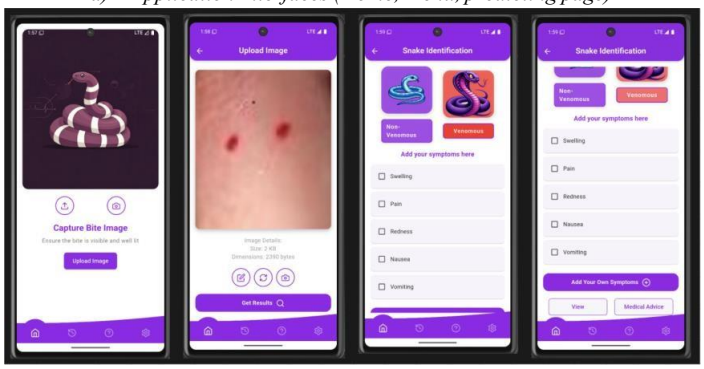
2) Seek more evaluation metrics and undertake tests with real-life examples.

3) Associate with end-user feedback to improve usability and functionability of the application.

D. *Future Direction*

Future studies can make snake identification from families more precise using bite marks analysis features. The incorporation of symptoms into the system will enhance the prediction; certain symptoms are observed to associate with particular snake families, apart from the venomous or non-venomous classifications. Measuring or analyzing the distance between the bite marks could give additional information on the species, while the size and shape of the marks also aid in identifying them. Heat signature scanning on the bite site may also reveal unique temperature patterns from different snake species which promise to be one of the good ways of enhancing the accuracy of the system.

a) Application Interfaces (Home, Menu, predicting page)



b) Application Interfaces (History, Support, Setting page)

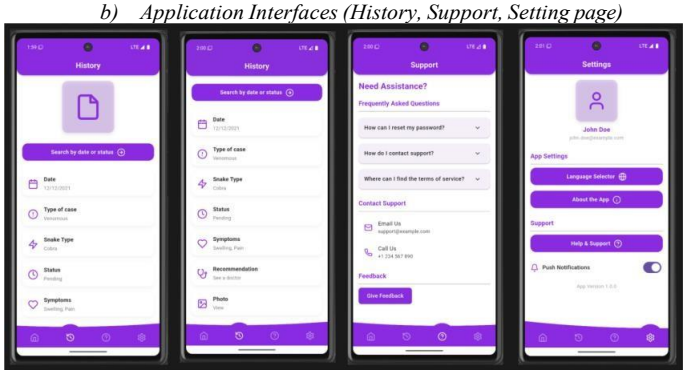


Fig. 10. Mobile application Interfaces

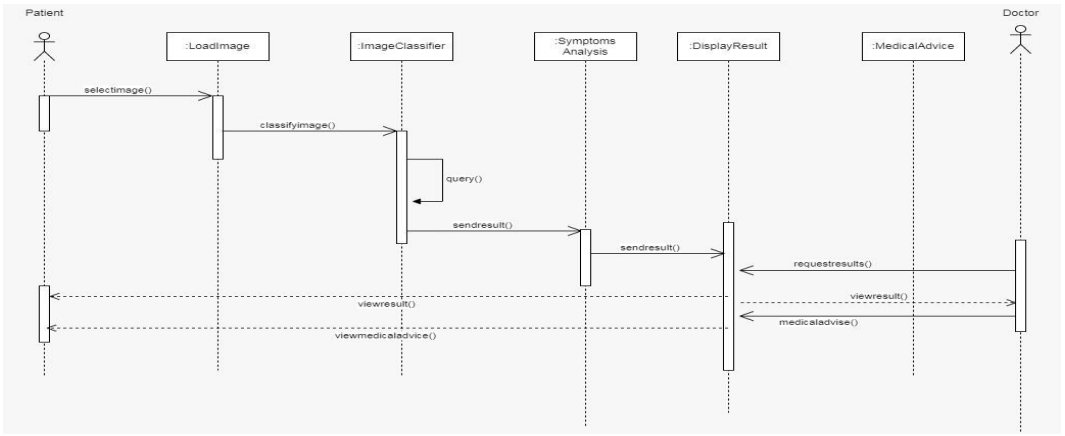


Fig. 11. Sequence Diagram

VI. CONCLUSION

Sri Lanka is home to numerous toxic and harmless snakes, and the purpose of this study is to design a mobile application that can identify the type of snake based on bite marks on the victim's skin in Sri Lanka. The main issue was the difficulty that doctors face when trying to identify the types of snakes, which is very important for choosing the right antivenom and effective treatment. Misidentification can increase the risks to the patients. The research started by understanding the challenges in identifying venomous and non-venomous snake bite marks. Machine learning methods were included for classification of images. Among these methods, transfer learning was selected for use with the MobileNetV2 deep convolutional neural network, since it works well for image tasks. The study indicates progress in computer vision using deep neural networks for image classification efficiently and accurately. This study provided a viable means for recognizing a snake species using transfer education. The developed mobile application can be used to classify whether a snake is venomous or not by capturing images of bite marks and symptoms of victimized individuals. The system resulted in output accuracy concerning predicted species and types of bite on the images given by users. The results were saved in a cloud database, allowing easy access to information and medical guidance. Further improvements, including more training with real bite marks, could enhance the model's reliability and usability. This development could turn into a helpful in mobile healthcare, aiding in snakebite identification and treatment choices.

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REFERENCES

[77] A. A. Awan, "A complete guide to data augmentation," DataCamp, Nov. 2022. [Online]. Available: https://www.datacamp.com/tutorial/complete-guide-data-augmentation.

[78] A. Afroz, B. N. Siddiquea, A. N. Shetty, T. N. W. Jackson, and A. D. Watt, "Assessing knowledge and awareness regarding snakebite and management of snakebite envenoming in healthcare workers and the general population: A systematic review and meta-analysis," *PLOS Neglected Tropical Diseases*, vol. 17, no. 2, p. e0011048, Feb. 2023, doi: 10.1371/journal.pntd.0011048.

[79] O. Ali, W. Abdelbaki, A. Shrestha, E. Elbasi, M. A. A. Alryalat, and Y. K. Dwivedi, "A systematic literature review of artificial intelligence in the healthcare sector: Benefits, challenges, methodologies, and functionalities," *Journal of Innovation & Knowledge*, vol. 8, no. 1, p. 100333, Jan. 2023, doi: 10.1016/j.jik.2023.100333.

[80] M. Bhardwaj, A. Singh, M. Kumar, S. K. Sharma, and J. Kumar, "Design and analysis of artificial intelligence model for the global issue of poisonous reptile identification," *BioGecko*, vol. 12, no. 4, pp. 16–25, Jul. 2023. [Online]. Available: https://biogecko.co.nz/index.php/journal/article/view/2

[81] H. K. Dishar and L. A. Muhammed, "A review of the overfitting problem in convolution neural network and remedy approaches," *Magallaẗ al-Qādisiyyaẗ li-ʿUlūm al-Ḥāsibāt wa-al-Riyāḍiyyāt*, vol. 15, no. 2, Sep. 2023, doi: 10.29304/jqcm.2023.15.2.1240.

[82] D. S. Ediriweera *et al.*, "Mapping the risk of snakebite in Sri Lanka: A national survey with geospatial analysis," *PLOS Neglected Tropical Diseases*, vol. 10, no. 7, pp. e0004813, Jul. 2016, doi: 10.1371/journal.pntd.0004813.

[83] "Deep learning model for identifying snakes by using snakes' bite marks," *IEEE Xplore*, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9104200

[84] N. P. A. U. D. Hernawati, N. Adiwijaya, and D. Q. Utama, "Image processing for snake identification based on bite using Local Binary Pattern and Support Vector Machine method," *Journal of Physics: Conference Series*, vol. 1192, pp. 012007, Mar. 2019, doi: 10.1088/1742-6596/1192/1/012007.